Governance memo GM-14: EA-IRMS normalization, acceptance and replicate screening

Issued by the facility quality board. Tables marked IN FORCE govern every batch recomputation.

Table N-1 is kept for audit history only; batches use the table that follows it.

Table N-1: Normalization reference sets [WITHDRAWN 2026-03-01]

element	reference_set	reject_limit_permil	max_rejections	slope_min	slope_max
N	RM-GLU-L;RM-CAF	0.40	3	0.900	1.100
C	RM-GLU-H;RM-CAF	0.35	2	0.900	1.100

Table N-2: Normalization reference sets [IN FORCE]

element	reference_set	reject_limit_permil	max_rejections	slope_min	slope_max
N	RM-GLU-L;RM-GLU-H	0.30	2	0.950	1.050
C	RM-GLU-L;RM-GLU-H;RM-CAF	0.25	3	0.950	1.050

Table A-1: Amplitude windows and blank limits [IN FORCE]

element	amplitude_min_mv	amplitude_max_mv	blank_area_max_vs
N	600	9000	0.3000
C	1000	16000	0.9000

Table L-1: Linearity application thresholds [IN FORCE]

element	threshold_permil_per_vs
N	0.0040
C	0.0025

Table Q-0: QC acceptance tolerances [SUPERSEDED]

material	element	tolerance_permil
QC-SED	N	0.30
QC-SED	C	0.25
QC-MUS	N	0.30
QC-MUS	C	0.25

Table Q-1: QC acceptance tolerances [IN FORCE]

material	element	tolerance_permil
QC-SED	N	0.25
QC-SED	C	0.20
QC-MUS	N	0.20
QC-MUS	C	0.15
QC-ALG	N	0.30
QC-ALG	C	0.25

Table rows above apply as printed.

Table X-1: Capsule flag codes [IN FORCE]

code	description	effect
F-DROP	capsule dropped during loading	EXCLUDE
F-TEAR	tin capsule torn	EXCLUDE
F-ASH	incomplete combustion residue	EXCLUDE
F-WET	sample visibly wet	EXCLUDE
F-SWAP	suspected capsule swap	EXCLUDE
F-NOTE	free text note	KEEP
F-LATE	weighed after cutoff	KEEP
F-LOWV	low amplitude noted	KEEP
F-REWT	reweighed once	KEEP
F-DUST	dust on tray noted	KEEP
F-STAT	static on balance	KEEP
F-COLR	unusual colour	KEEP
F-SMLL	small capsule fold	KEEP
F-OLDB	old batch label	KEEP
F-RECK	rechecked by second analyst	KEEP
F-HUMD	humid day note	KEEP

Table C-1: Reintegration reason codes [IN FORCE]

code	description	status
RI-01	baseline drift under the peak	ACCEPTED
RI-02	peak tailing truncated	ACCEPTED
RI-03	split peak merged	ACCEPTED
RI-04	shoulder from previous peak	ACCEPTED
RI-05	start point moved to valley	ACCEPTED
RI-06	end point moved to baseline	ACCEPTED
RI-07	analyst preference	REJECTED
RI-08	value looked unexpected	REJECTED
RI-09	match published value	REJECTED
RI-10	unrecorded reason	REJECTED
RI-11	noise spike removed	ACCEPTED
RI-12	reference pulse overlap	ACCEPTED

Table C-1 (continued)

code	description	status
RI-13	smoothing changed	REJECTED
RI-14	threshold lowered to include peak	REJECTED
RI-15	background subtraction window moved	ACCEPTED
RI-16	integration batch default	REJECTED
RI-17	carry-over correction	REJECTED
RI-18	detector saturation flag cleared	REJECTED
RI-19	manual baseline from blank	ACCEPTED
RI-20	duplicate peak removed	ACCEPTED

Table G-0: Grubbs critical values, one-sided [SUPERSEDED]

n	alpha_0.05	alpha_0.01
3	1.1531	1.1546
4	1.4625	1.4925
5	1.6714	1.7489
6	1.8221	1.9442
7	1.9381	2.0973
8	2.0317	2.2208
9	2.1096	2.3231
10	2.1761	2.4097
11	2.2339	2.4843
12	2.2850	2.5494
13	2.3305	2.6070
14	2.3717	2.6585
15	2.4090	2.7049
16	2.4433	2.7470
17	2.4748	2.7854
18	2.5040	2.8208
19	2.5312	2.8535
20	2.5566	2.8838
21	2.5804	2.9121
22	2.6028	2.9385
23	2.6239	2.9633
24	2.6439	2.9866
25	2.6629	3.0086
26	2.6809	3.0295
27	2.6981	3.0492
28	2.7145	3.0680
29	2.7301	3.0859
30	2.7451	3.1029
31	2.7595	3.1192
32	2.7733	3.1348
33	2.7866	3.1497
34	2.7994	3.1640
35	2.8118	3.1778
36	2.8237	3.1910
37	2.8352	3.2038
38	2.8463	3.2161
39	2.8571	3.2280
40	2.8675	3.2395
41	2.8777	3.2506
42	2.8875	3.2613
43	2.8970	3.2717
44	2.9063	3.2818
45	2.9153	3.2916
46	2.9241	3.3011
47	2.9326	3.3104
48	2.9409	3.3194
49	2.9491	3.3281
50	2.9570	3.3366
51	2.9647	3.3449
52	2.9722	3.3530
53	2.9796	3.3609
54	2.9868	3.3686
55	2.9938	3.3761
56	3.0007	3.3834
57	3.0075	3.3905
58	3.0141	3.3975
59	3.0205	3.4044
60	3.0269	3.4111

Table G-1: Grubbs critical values, two-sided [IN FORCE]

n	alpha_0.05	alpha_0.01
3	1.1543	1.1547
4	1.4812	1.4962
5	1.7150	1.7637
6	1.8871	1.9728
7	2.0200	2.1391
8	2.1266	2.2744
9	2.2150	2.3868
10	2.2900	2.4821
11	2.3547	2.5641
12	2.4116	2.6357
13	2.4620	2.6990
14	2.5073	2.7554
15	2.5483	2.8061
16	2.5857	2.8521
17	2.6200	2.8940
18	2.6516	2.9325
19	2.6809	2.9680
20	2.7082	3.0008
21	2.7338	3.0314
22	2.7577	3.0599
23	2.7803	3.0866
24	2.8016	3.1117
25	2.8217	3.1353
26	2.8408	3.1577
27	2.8589	3.1788
28	2.8762	3.1989
29	2.8927	3.2179
30	2.9085	3.2361
31	2.9236	3.2534
32	2.9380	3.2700
33	2.9519	3.2858
34	2.9653	3.3010
35	2.9782	3.3156
36	2.9906	3.3296
37	3.0026	3.3431
38	3.0141	3.3561
39	3.0253	3.3686
40	3.0361	3.3807
41	3.0466	3.3924
42	3.0567	3.4037
43	3.0666	3.4146
44	3.0761	3.4252
45	3.0854	3.4354
46	3.0945	3.4454
47	3.1032	3.4551
48	3.1118	3.4645
49	3.1201	3.4736
50	3.1282	3.4825
51	3.1362	3.4911
52	3.1439	3.4995
53	3.1514	3.5077
54	3.1588	3.5157
55	3.1660	3.5235
56	3.1730	3.5311
57	3.1799	3.5386
58	3.1866	3.5458
59	3.1932	3.5529
60	3.1997	3.5598
61	3.2060	3.5666
62	3.2122	3.5733
63	3.2182	3.5798
64	3.2242	3.5861
65	3.2300	3.5924
66	3.2357	3.5985
67	3.2413	3.6044
68	3.2469	3.6103
69	3.2523	3.6161
70	3.2576	3.6217
71	3.2628	3.6272
72	3.2680	3.6327

Table G-1 (continued)

n	alpha_0.05	alpha_0.01
73	3.2730	3.6380
74	3.2780	3.6433
75	3.2829	3.6484
76	3.2877	3.6535
77	3.2924	3.6585
78	3.2970	3.6633
79	3.3016	3.6682
80	3.3061	3.6729
81	3.3106	3.6775
82	3.3149	3.6821
83	3.3192	3.6866
84	3.3235	3.6911
85	3.3277	3.6955
86	3.3318	3.6998
87	3.3359	3.7040
88	3.3399	3.7082
89	3.3438	3.7123
90	3.3477	3.7164
91	3.3516	3.7204
92	3.3554	3.7243
93	3.3591	3.7282
94	3.3628	3.7320
95	3.3665	3.7358
96	3.3701	3.7396
97	3.3737	3.7432
98	3.3772	3.7469
99	3.3807	3.7505
100	3.3841	3.7540
101	3.3875	3.7575
102	3.3908	3.7610
103	3.3941	3.7644
104	3.3974	3.7677
105	3.4006	3.7711
106	3.4038	3.7744
107	3.4070	3.7776
108	3.4101	3.7808
109	3.4132	3.7840
110	3.4163	3.7871
111	3.4193	3.7902
112	3.4223	3.7933
113	3.4253	3.7963
114	3.4282	3.7993
115	3.4311	3.8023
116	3.4340	3.8052
117	3.4368	3.8081
118	3.4396	3.8110
119	3.4424	3.8138
120	3.4451	3.8166
121	3.4479	3.8194
122	3.4506	3.8222
123	3.4533	3.8249
124	3.4559	3.8276
125	3.4585	3.8302
126	3.4611	3.8329
127	3.4637	3.8355
128	3.4662	3.8381
129	3.4688	3.8406
130	3.4713	3.8432
131	3.4738	3.8457
132	3.4762	3.8482
133	3.4786	3.8507
134	3.4811	3.8531
135	3.4835	3.8555
136	3.4858	3.8579
137	3.4882	3.8603
138	3.4905	3.8626
139	3.4928	3.8650
140	3.4951	3.8673

Table R-1: Session rulings [IN FORCE]

ruling	date	project	target	element	action	status
RL-0340	2025-09-28	HSI-2581	SQ-2581-01	ALL	EXCLUDE_POSITION	SUPERSEDED
RL-0455	2025-10-12	HSI-2581	SQ-2577-01	C	EXCLUDE_POSITION	SUPERSEDED
RL-0351	2025-10-16	HSI-2598	SQ-2581-26	N	ACCEPT_SESSION	SUPERSEDED
RL-0385	2025-10-25	HSI-2598	SQ-2581-14	C	EXCLUDE_POSITION	RESCINDED
RL-0443	2025-10-26	HSI-2577	SQ-2577-27	ALL	ACCEPT_SESSION	IN FORCE
RL-0377	2025-10-29	HSI-2577	SQ-2581-26	C	EXCLUDE_POSITION	RESCINDED
RL-0348	2025-11-02	HSI-2581	SQ-2577-21	N	REJECT_SESSION	IN FORCE
RL-0446	2025-11-06	HSI-2598	SQ-2598-29	ALL	EXCLUDE_POSITION	RESCINDED
RL-0343	2025-11-15	HSI-2581	SQ-2581-20	N	REJECT_SESSION	RESCINDED
RL-0372	2025-11-24	HSI-2598	SQ-2598-21	C	ACCEPT_SESSION	IN FORCE
RL-0469	2025-11-25	HSI-2598	SQ-2598-28	ALL	ACCEPT_SESSION	SUPERSEDED
RL-0395	2025-12-07	HSI-2581	SQ-2577-21	C	EXCLUDE_POSITION	SUPERSEDED
RL-0429	2025-12-10	HSI-2581	SQ-2577-29	C	ACCEPT_SESSION	SUPERSEDED
RL-0380	2025-12-18	HSI-2577	SQ-2577-18	ALL	EXCLUDE_POSITION	SUPERSEDED
RL-0436	2025-12-24	HSI-2577	SQ-2598-08	ALL	REJECT_SESSION	IN FORCE
RL-0404	2026-01-02	HSI-2581	SQ-2577-09	C	EXCLUDE_POSITION	SUPERSEDED
RL-0369	2026-01-05	HSI-2581	SQ-2598-01	N	EXCLUDE_POSITION	RESCINDED
RL-0396	2026-01-09	HSI-2577	SQ-2577-19	N	ACCEPT_SESSION	IN FORCE
RL-0397	2026-01-13	HSI-2598	SQ-2598-09	ALL	REJECT_SESSION	RESCINDED
RL-0401	2026-02-01	HSI-2577	SQ-2598-20	N	EXCLUDE_POSITION	RESCINDED
RL-0402	2026-02-06	HSI-2598	SQ-2581-11	C	EXCLUDE_POSITION	IN FORCE
RL-0413	2026-02-09	HSI-2581	SQ-2598-25	ALL	ACCEPT_SESSION	IN FORCE
RL-0418	2026-02-13	HSI-2581	SQ-2581-12	N	REJECT_SESSION	SUPERSEDED
RL-0360	2026-02-27	HSI-2581	SQ-2598-30	ALL	REJECT_SESSION	RESCINDED
RL-0389	2026-03-07	HSI-2577	SQ-2598-08	C	EXCLUDE_POSITION	IN FORCE
RL-0410	2026-03-11	HSI-2581	SQ-2598-14	C	ACCEPT_SESSION	SUPERSEDED
RL-0442	2026-03-23	HSI-2598	SQ-2581-13	ALL	ACCEPT_SESSION	SUPERSEDED
RL-0398	2026-04-07	HSI-2577	SQ-2577-24	C	EXCLUDE_POSITION	SUPERSEDED
RL-0335	2026-04-13	HSI-2598	SQ-2598-17	C	REJECT_SESSION	IN FORCE
RL-0415	2026-04-30	HSI-2581	SQ-2577-14	C	REJECT_SESSION	IN FORCE
RL-0353	2026-05-04	HSI-2598	SQ-2598-28	C	ACCEPT_SESSION	RESCINDED
RL-0423	2026-05-05	HSI-2581	SQ-2581-05	ALL	REJECT_SESSION	SUPERSEDED
RL-0375	2026-05-11	HSI-2598	SQ-2581-16	ALL	EXCLUDE_POSITION	RESCINDED
RL-0333	2026-05-13	HSI-2598	SQ-2581-11	N	ACCEPT_SESSION	RESCINDED
RL-0450	2026-05-31	HSI-2577	SQ-2581-17	C	ACCEPT_SESSION	SUPERSEDED
RL-0470	2026-06-04	HSI-2581	SQ-2581-25	C	REJECT_SESSION	RESCINDED
RL-0426	2026-06-07	HSI-2577	SQ-2577-27	C	REJECT_SESSION	SUPERSEDED
RL-0460	2026-06-08	HSI-2581	SQ-2577-20	C	EXCLUDE_POSITION	SUPERSEDED
RL-0452	2026-06-09	HSI-2581	SQ-2581-07	ALL	ACCEPT_SESSION	SUPERSEDED
RL-0346	2026-06-13	HSI-2598	SQ-2581-03	ALL	ACCEPT_SESSION	RESCINDED
RL-0338	2026-06-17	HSI-2577	SQ-2581-03	C	REJECT_SESSION	IN FORCE
RL-0392	2026-06-22	HSI-2598	SQ-2581-06	ALL	REJECT_SESSION	IN FORCE
RL-0361	2026-06-23	HSI-2581	SQ-2581-12	C	ACCEPT_SESSION	SUPERSEDED
RL-0368	2026-06-25	HSI-2581	SQ-2581-28	N	ACCEPT_SESSION	IN FORCE
RL-0417	2026-07-02	HSI-2577	SQ-2577-14	ALL	REJECT_SESSION	IN FORCE
RL-0358	2026-07-05	HSI-2598	SQ-2577-06	C	EXCLUDE_POSITION	RESCINDED
RL-0462	2026-07-08	HSI-2577	SQ-2577-25	N	ACCEPT_SESSION	RESCINDED
RL-0458	2026-07-09	HSI-2577	SQ-2598-25	N	EXCLUDE_POSITION	IN FORCE
RL-0473	2026-07-09	HSI-2598	SQ-2598-04	N	EXCLUDE_POSITION	RESCINDED
RL-0447	2026-07-11	HSI-2581	SQ-2581-07	C	EXCLUDE_POSITION	IN FORCE
RL-0466	2026-07-13	HSI-2577	SQ-2577-22	ALL	ACCEPT_SESSION	IN FORCE
RL-0407	2026-07-20	HSI-2577	SQ-2598-17	N	REJECT_SESSION	SUPERSEDED
RL-0432	2026-07-20	HSI-2598	SQ-2581-02	ALL	EXCLUDE_POSITION	RESCINDED
RL-0329	2026-07-31	HSI-2577	SQ-2598-11	N	REJECT_SESSION	SUPERSEDED
RL-0339	2026-08-05	HSI-2598	SQ-2581-08	N	ACCEPT_SESSION	SUPERSEDED
RL-0439	2026-08-22	HSI-2581	SQ-2598-28	N	REJECT_SESSION	IN FORCE
RL-0421	2026-08-23	HSI-2598	SQ-2581-12	N	EXCLUDE_POSITION	RESCINDED
RL-0465	2026-08-24	HSI-2581	SQ-2577-22	C	REJECT_SESSION	RESCINDED
RL-0433	2026-08-30	HSI-2581	SQ-2577-11	C	ACCEPT_SESSION	SUPERSEDED
RL-0327	2026-09-01	HSI-2581	SQ-2581-09	C	EXCLUDE_POSITION	SUPERSEDED
RL-0336	2026-09-02	HSI-2581	SQ-2581-23	C	EXCLUDE_POSITION	RESCINDED
RL-0383	2026-09-02	HSI-2577	SQ-2581-29	C	REJECT_SESSION	SUPERSEDED
RL-0386	2026-09-05	HSI-2581	SQ-2577-21	C	REJECT_SESSION	SUPERSEDED
RL-0445	2026-09-06	HSI-2577	SQ-2577-14	ALL	REJECT_SESSION	IN FORCE
RL-0356	2026-09-08	HSI-2581	SQ-2581-15	ALL	EXCLUDE_POSITION	RESCINDED
RL-0355	2026-09-09	HSI-2581	SQ-2598-24	C	REJECT_SESSION	IN FORCE
RL-0366	2026-09-10	HSI-2577	SQ-2577-18	N	ACCEPT_SESSION	SUPERSEDED
RL-0364	2026-09-11	HSI-2581	SQ-2581-16	N	ACCEPT_SESSION	RESCINDED
RL-0332	2026-09-14	HSI-2577	SQ-2581-30	ALL	REJECT_SESSION	IN FORCE
RL-0384	2026-09-14	HSI-2581	SQ-2577-24	ALL	REJECT_SESSION	RESCINDED

Table R-1 (continued)

ruling	date	project	target	element	action	status
RL-0475	2026-11-07	HSI-2611	SQ-2611-05	C	ACCEPT_SESSION	IN FORCE
RL-0478	2026-11-07	HSI-2611	SQ-2611-13	N	REJECT_SESSION	IN FORCE
RL-0480	2026-11-05	HSI-2611	SQ-2611-18	ALL	REJECT_SESSION	RESCINDED
RL-0482	2026-11-06	HSI-2611	SQ-2611-16	ALL	REJECT_SESSION	IN FORCE
RL-0484	2026-11-09	HSI-2611	SQ-2611-16	ALL	ACCEPT_SESSION	IN FORCE
RL-0486	2026-11-07	HSI-2598	SQ-02	ALL	ACCEPT_SESSION	IN FORCE
RL-0489	2026-11-08	HSI-2611	SQ-07/27	ALL	EXCLUDE_POSITION	IN FORCE
RL-0491	2026-11-08	HSI-2611	2611/20/31	ALL	EXCLUDE_POSITION	IN FORCE
RL-0492	2026-11-08	HSI-2611	sq-2611-13/11	ALL	EXCLUDE_POSITION	IN FORCE
RL-0494	2026-11-08	HSI-2611	RUN-K1-1101/42	ALL	EXCLUDE_POSITION	RESCINDED